HCF & LCM

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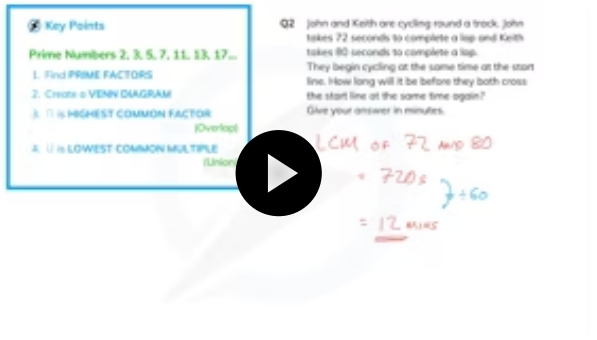

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Highest common factor (HCF)

What is the highest common factor (HCF) of two numbers?

- A **common factor** of two numbers is a value that **both numbers** can be divided by, leaving no remainder
  - **1** is always a common factor of any two numbers
  - Any **factor** of a **common factor** will also be a common factor of the original two numbers
    - 6 is a common factor of 24 and 30
    - Therefore 1, 2 and 3 are also common factors of 24 and 30
- The **highest** common factor is the largest common factor of the two numbers
  - The highest common factor is useful when simplifying fractions or factorising expressions

How do I find the highest common factor (HCF) of two numbers?

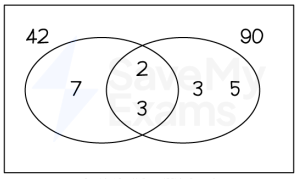
- To find **common factors**:
  - write out the **factors** of each number in a list
  - identify the numbers that appear in **both** lists

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How can I use a Venn diagram to find the highest common factor (HCF) of two numbers?

- Write each number as a **product of its prime factors**
  - 42 = 2×3×7 and 90 = 2×3×3×5
- Find the prime factors that are **common** to **both numbers** and put these in the **centre of the Venn diagram**
  - 42 and 90 both have a prime factor of 2
    - Put 2 in the centre of the diagram
  - Although 3 appears twice in the prime factors of 90, it appears once in the prime factors of 42
    - Put a single 3 in the centre of the diagram
  - If there are no common prime factors, put a 1 in the centre of the diagram
- Put the **remaining prime factors** in the respective regions
  - 7 would go in the region for 42
  - 3 and 5 would go in the region for 90
- The **highest common factor** is the **product** of the **numbers in the centre**
  - The HCF of 42 and 90 is 2×3, which is 6
- If there are no common prime factors then the HCF is 1



How can I use the powers of prime factors to find the highest common factor (HCF) of two numbers?

- Write each number as a **product of the powers of its prime factors**
  - 24 = 2^3×3 and 60 = 2^2×3×5
- Find all **common** prime factors and identify the **highest power** that appears in both numbers
  - The highest power of 2 in both is 2^2
    - 2^2 is a common factor
  - The highest power of 3 in both is 3^1
    - 3 is a common factor
  - No other prime number appears in both
- The **highest common factor** is the **product** of the common powers of primes
  - The HCF of 24 and 60 is 2^2×3 which is 12

The highest common factor of two numbers could be one of the numbers!

E.g. The highest common factor of 4 and 12 is 4.

Worked Example

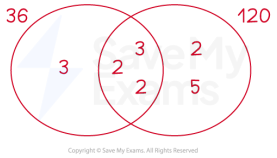
Find the highest common factor of 36 and 120.

Answer:

Write both numbers as a product of prime factors

$$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$
$$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3 \times 5$$

Write the prime factors in a Venn diagram



Multiply the common prime factors in the centre

$$\text{HCF} = 2 \times 2 \times 3$$

Alternatively, list the factors for each number

36: 1, 2, 3, 4, 6, 9, <u>12</u>, 18, 36
120: 1, 2, 3, 4, 5, 6, 8, 10, <u>12</u>, 15, 20, 24, 30, 40, 60, 120

Another alternative is to find the highest common powers of primes

2^2 and 3^1 are the highest common powers of primes
$\text{HCF} = 2^2 \times 3^1$

HCF = 12

Lowest common multiple (LCM)

What is the lowest common multiple (LCM) of two numbers?

- A **common multiple** of two numbers is a number that appears in **both** of their times tables
  - The **product** of the original two numbers is always a common multiple (but not necessarily the lowest)
  - **Any multiple** of a **common multiple** will **also be a common multiple** of the original two numbers
    - 30 is a common multiple of 3 and 10

multiple between two numbers
    - This is useful when finding a common denominator and when adding or subtracting fractions

How do I find the lowest common multiple (LCM) of two numbers?

- To find the **lowest common multiple** of two numbers:
    - write out the first few **multiples** of each number
    - identify the multiples that appear in **both** lists
        - If there are **none** then write out the **next few multiples** of each number until you find a common multiple
- The **lowest common multiple** will be the **smallest** multiple that appears in **both** lists

How can I use a Venn diagram to find the lowest common multiple (LCM) of two numbers?

- Write each number as a **product of its prime factors**
    - 42 = 2×3×7 and 90 = 2×3×3×5
- Find the prime factors that are **common** to **both numbers** and put these in the **centre of the Venn diagram**
    - 42 and 90 both have a prime factor of 2
        - Put a 2 in the centre of the diagram
    - Although 3 appears twice in the prime factors of 90, it appears once in the prime factors of 42
        - Put a single 3 in the centre of the diagram
    - If there are no common prime factors then put a 1 in the centre of the diagram
- Put the **remaining prime factors** in the respective regions
    - 7 would go in the region for 42
    - 3 and 5 would go in the region for 90
- The **lowest common multiple** is the **product** of **all the numbers in the circles**
    - The LCM of 42 and 90 is 7×2×3×3×5, which is 630

How can I use the powers of prime factors to find the lowest common multiple (LCM) of two numbers?

- Write each number as a **product of the powers of its prime factors**
    - $72 = 2^3 \times 3^2$ and $540 = 2^2 \times 3^3 \times 5$

- 2^3 is the highest power of 2 shown (from 72)
- 3^3 is the highest power of 3 shown (from 540)
- 5^1 is the highest power of 5 shown (from 540)
  - Note: 5 is not a common prime in 72 and 540, but it is still needed for the LCM

- The **lowest common multiple** is the **product** of these **highest powers**
  - The LCM of 72 and 540 is $2^3 \times 3^3 \times 5$, which is 1080

Examiner Tips and Tricks

The lowest common multiple of two numbers could be one of the numbers!

E.g. The lowest common multiple of 4 and 12 is 12.

Worked Example

Find the lowest common multiple of 36 and 120.

Answer:

Write both numbers as a product of prime factors

$$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$$
$$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3 \times 5$$

Write the prime factors in a Venn diagram

Multiply all the prime factors in the diagram

$$\text{LCM} = 3 \times 2 \times 2 \times 3 \times 2 \times 5$$

An alternative method is to write out the multiples

36: 26, 72, 108, 144, 180, 216, 252, 288, 324, <u>360</u>, 396, ...
120: 120, 240, <u>360</u>, 480, ...

Another alternative method is to find the highest powers of each and every prime that appears
Then multiply these together

$$2^3 \times 3^2 \times 5^1$$

LCM = 360

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